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Ans 1: CPU registers are small, high-speed storage locations inside the processors.

Importance:

- Store intermediate data and instructions
- Faster than cache and memory
- Improve execution speed of programs

Ans 2: Program Counter (PC) is a register that holds the address of the next instruction.

Function:

- Points to next instruction to be executed
- Automatically updates after each instruction
- Controls instruction sequence.

Ans 3: Instruction format defines the structure of an instruction in binary form.

Key Fields:

- Opcode (operation)
- Operands
- Addressing mode

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Ans 5: Addressing mode specify how to locate operands.

Purpose:

- Provide flexibility in accessing data
- Reduce instruction count
- Improve program efficiency

Ans 6: ISA is the interface between hardware and software defining supported instructions.

Includes:

- Instruction formats
- Data types
- Registers and addressing modes

Ans 7: Types of instructions in a processor are:

- Arithmetic (ADD, SUB)
- Logical (AND, OR)
- Data Transfer (LOAD, STORE)
- Control (JUMP, BRANCH)

Ans 8: Explanation:

- Opcode is interpreted
- Required operation is identified
- Control signals are generated
- Operands are prepared for execution

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Ans 9: Basic instruction cycle are:

- Fetch instruction from memory
- Decode instruction
- execute operation
- Store result

Ans 10: Characteristics of RISC processors are:

- Simple and fixed-length instructions
- few addressing modes
- Faster execution (pipeline-friendly)
- Load/store architecture

Ans 11: Characteristics of CISC processors are:

- Complex and variable-length instructions
- Many addressing modes
- fewer instructions per program
- Slower but powerful instructions.

Ans 12: RISC-V:

↳ Open-source ISA
↳ Simple, fixed-length instructions
↳ Easier to design and extend

x86:

↳ Proprietary (Intel/AMD)
↳ Complex, variable-length instructions

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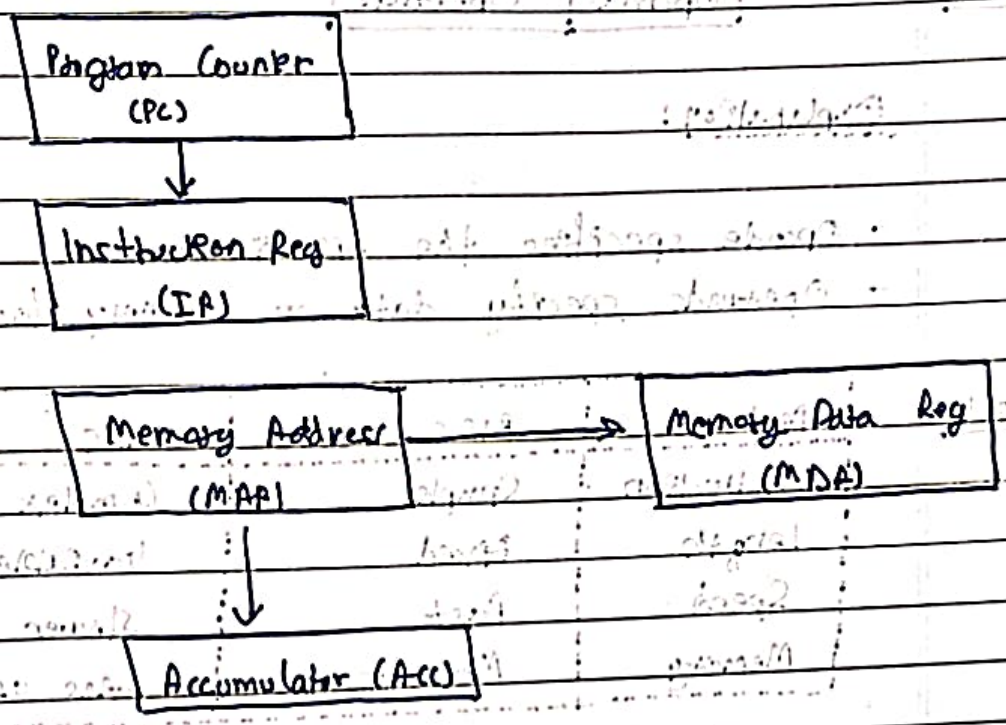
* Section - B

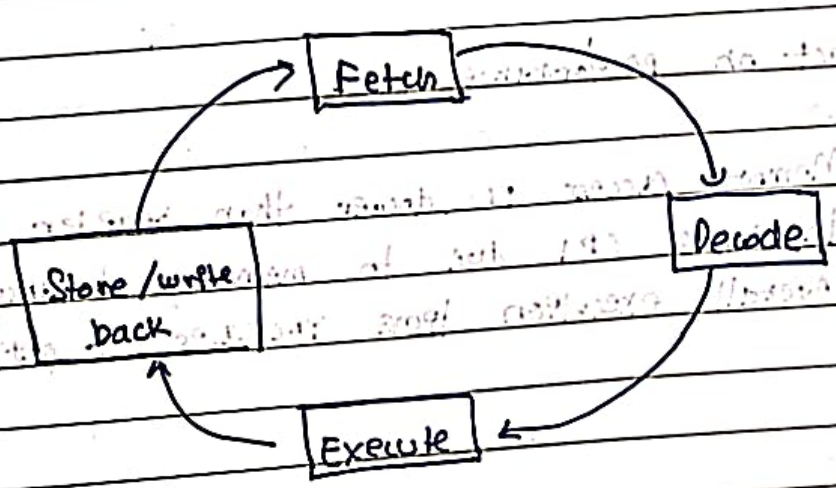
Ans 1:



Explanation:

- PC holds next instruction address; IR stores current instruction
- MAR & MDR handle memory access; Acc. stores intermediate results.

Ans 2:



- CPU fetches instructions, decode it, execute it and write back.

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Ans 3:

Opcode	Operands

Explanation:

- Opcode specifies the operations
- Operands specify data or memory location.

Ans 4:

Feature	RISC	CISC
Instruction	Simple	Complex
Length	Fixed	Variable
Speed	Fast	Slower
Memory	More usage	Less usage

Explanation:

- RISC focuses on simplicity and speed.
- CISC focuses on powerful and complexity.

* Section - C

Ans 1: Impact on performance are:

→ Memory access is slower than register access
→ Increases CPI due to memory latency
→ Overall execution time increases → performance decrease

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Ans 2: The should choose RISC architecture because:

→ Simple and fixed-length instructions
→ Easier to pipeline efficiently
→ Requires simpler hardware design

Ans 3: CPU executes this instruction as:

Step 1: Fetch
- Instruction is fetched from memory onto IR

Step 2: Decode
- Opcode is decoded and registers are identified

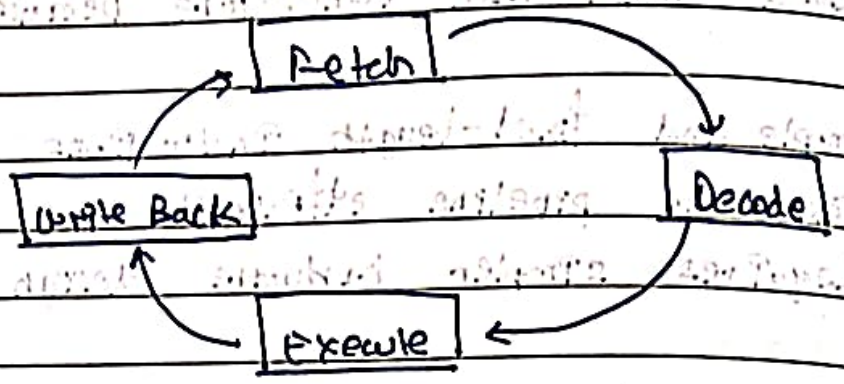
Step 3: Execute
- Data is read from registers
- ALU performs operation
- Result is stored back in register

Ans 4: x86 is widely used despite complexity because:

→ Backward compatibility with older software
→ Large ecosystem (OS, applications, industry support)
→ Modern CPUs internally optimized.
→ Despite complexity, compatibility and ecosystem keep x86 dominant

* Section - D

Ans 1:



Explanation of fetch stage:

- PC holds address of next instructions
- Address sent to memory via MAR
- Instruction fetched into MBR and loaded into IR
- PC is incremented to next instruction.

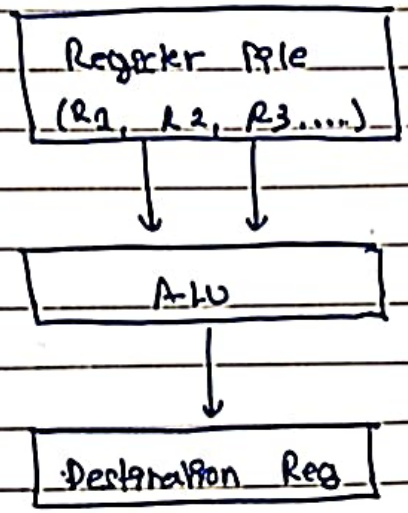
Ans 2:

funct 7	rs2	rs1	funct3	rd	opcode

Explanation:

→ opcode - Specifies operation type
→ rd - Destination register
→ rs1, rs2 - Source Register
→ funct3 & funct7 - Define exact operation
→ Fixed-length (32-bit) format simplifies decoding

Ans 3:



Explanation:

- Data is read from source registers.
- Sent to ALU for operation
- Result is written back to destination register
- Enable fast computation without memory access